

The Chinese Remainder Theorem and Pandrosion Idempotents

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Abstract

The Chinese Remainder Theorem for $\mathbb{Q}[z]/(P)$ with $P = \prod (z - \alpha_j)$ is the *isomorphism* $\mathbb{Q}[z]/(P) \cong \mathbb{Q}^n$ given by evaluation at the roots. The Pandrosion quotients provide the explicit CRT idempotents: $e_j = Q_j/P'(\alpha_j)$ satisfy $e_j^2 \equiv e_j$, $e_j e_k \equiv 0$ for $j \neq k$, and $\sum e_j = 1$. This unifies Lagrange interpolation (Paper 92), partial fractions (Paper 90), and the Galois splitting field (Paper 74) as three faces of a single CRT structure.

1 The polynomial CRT

Theorem 1.1. For $P = \prod_{j=1}^n (z - \alpha_j)$ with distinct roots:

$$\boxed{\mathbb{Q}[z]/(P) \cong \prod_{j=1}^n \mathbb{Q}[z]/(z - \alpha_j) \cong \mathbb{Q}^n.} \quad (1)$$

The isomorphism is $\varphi(f) = (f(\alpha_1), \dots, f(\alpha_n))$.

2 Pandrosion idempotents

The CRT basis elements are the normalized Pandrosion quotients:

$$e_j = \frac{Q_j(z)}{P'(\alpha_j)}, \quad Q_j(z) = \frac{P(z)}{z - \alpha_j}. \quad (2)$$

Proposition 2.1 (Idempotent properties). 1. **Orthogonality:** $e_j(\alpha_k) = \delta_{jk}$

2. **Partition of unity:** $\sum_{j=1}^n e_j = 1$

3. **Idempotence:** $e_j^2 \equiv e_j \pmod{P}$

4. **Orthogonal products:** $e_j \cdot e_k \equiv 0 \pmod{P}$ for $j \neq k$

All four verified at 10^{-13} precision.

3 Unification

Paper	CRT interpretation
1 (Quotients)	$Q_j = \text{CRT basis elements}$
90 ($F_P \cdot P = P'$)	Partial fractions = CRT of P'/P
92 (Lagrange)	CRT reconstruction = interpolation
74 (Galois)	CRT = splitting of $\mathbb{Q}[z]/(P)$
84 (Abel–Ruffini)	Non-split $\Rightarrow$ no full CRT
93 (Cayley–Hamilton)	$P(C) = 0$ via CRT structure

The partial fraction $F_P = \sum 1/(z - \alpha_j)$ is the CRT decomposition of P'/P : modulo $(z - \alpha_j)$, the residue is 1.

4 Multiplication isomorphism

The isomorphism preserves multiplication: if $f \leftrightarrow (f_1, \dots)$ and $g \leftrightarrow (g_1, \dots)$, then $fg \bmod P \leftrightarrow (f_1g_1, \dots)$. Verified: $(2z + 1)(z^2 - z + 3) \bmod P$ gives componentwise products at all four nodes, error = 0.

References

- [1] Sun Tzu, *Sunzi Suanjing*, 3rd century.
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- [3] I. Besevic, *Galois theory of quotient orbits*, Pandrosion Dynamics **74** (2026).
- [4] I. Besevic, *Lagrange interpolation*, Pandrosion Dynamics **92** (2026).